

Spontaneous Emergence of Communication in Evolved Foraging Agents

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Abstract—Can communication emerge through evolution of simple foraging agents as a side effect of selection, even when nothing in the fitness function ever rewards signalling? In this paper, we simulate a population of fixed-topology neural agents foraging a 2-D world where food is sensed only at short range but agents can emit a scalar signal heard at long range, so that a finder *could* recruit groupmates. Fitness only awards food eaten. Two controllers are studied: a memoryless feedforward network and an Elman recurrent network, both with three levels of selection: clonal *colony*, *individual*, and a 50/50 *hybrid*, all measured under a signal-ablation control that severs perception. In the end, results show that functional communication *does* emerge, but only in the feedforward memoryless agents: enabling the channel raises yield in all three regimes (colony, individual and hybrid). On the other hand, the recurrent agents forage substantially better overall yet derive no benefit from the channel, possibly because memory allows them to develop better foraging strategies that beat relying on other agents’ signals.

I. INTRODUCTION

The starting point for this project was one of the main curiosities that initially led me to pursue an MSc in this field: can the autonomous evolution of neural network agents, aimed at maximizing a given objective, lead to the emergence of behaviors and strategies unrelated to that objective as a side effect? This work tries to explore exactly this, in a Bio-Inspired Artificial Intelligence setting.

Specifically, the question is whether communication can arise spontaneously in a population of foraging agents, even though fitness never rewards it. Evolutionary robotics shows this can happen: Floreano *et al.* [1] and Mitri *et al.* [2] evolved foragers that signal food location when relatedness or colony-level selection aligned with individual and group interest. The point is therefore not to show again that a free channel can be created, but to ask what governs whether it is, and whether giving the agents memory influences it.

To isolate the first question, the task is kept minimal: food is detectable only at close range while signals travel far, so an agent that finds food can lead others to it only by signalling. With no poison to disambiguate and no explicit cost to signalling, the selection regime is essentially the only social pressure at play.

Within this frame, two factors are manipulated. The first is the *scope of the objective*, spanning from colony-level collective optimization, through an intermediate hybrid, to purely individual selection: the initial hypothesis is that group-level selection should favor cooperation and possibly evolve the use of the signal, and individual selection to neglect or abuse it. [1]

The second is *memory*: alongside the memoryless feedforward

neural network agents, agents are equipped with an Elman RNN. This was done with the objective of finding out whether the capacity to integrate signals over time can help communication emerge, under the hypothesis that agents able to remember past signals might develop richer communication. A difficulty intrinsic to this setting is measurement: if communication is never specified, how can we recognize it when it appears? This problem is addressed on two complementary fronts. To test whether the channel is *functionally* useful, every condition is re-evolved with signal perception severed, i.e. a *deaf* control (exploiting the idea that lack of reception effectively equals lack of communication), and the resulting change in yield is measured, isolating what communication is actually worth.

To characterize what the signal *carries*, the mutual information between an agent’s emitted signal and different environment variables (its own food state, and the spatial configuration of its neighbours) is measured. This is also done against the deaf baseline, so that information arising merely because the signal shares network inputs with the sensors is not mistaken for genuine communication.

Together we believe these give a functional and an informational view of communication emergence.

II. TASK AND ENVIRONMENT

World. $G = 8$ agents forage a square torus, a shape chosen to avoid behavior around the borders to be a factor, for $T = 150$ steps. Food is clustered in $P = 2$ *renewable* patches of 12 items; an eaten item respawns at its patch after a short delay. *Renewability* (see Sec. VII) makes a patch a persistent but hard-to-find resource, so the binding constraint on yield is discovery and exploitation effort rather than a finite pile: the regime in which recruitment can pay.

Sensing, acting, signalling. Food is sensed only within $r_{\text{sense}}=0.05$ and eaten only within $r_{\text{eat}}=0.03$; each agent emits a scalar $s \in [0, 1]$ heard by groupmates within $r_{\text{signal}}=0.55$.

This range asymmetry is the reason communication could help. Eating is rate-limited to one item per agent per step, so a lone finder cannot instantly drain a patch; instead, directing idle groupmates onto known food can raise the group’s harvest.

The 9-D observation consists of four distance-weighted directional food sensors, an on-food bit, and four directional signal sensors aggregating groupmates’ emissions by bearing (an agent never hears itself). Outputs are a 2-D move and the emitted signal.

Crucially, **nothing in the fitness function references s .**

Ablation control. To measure the channel’s *functional* value

every condition is re-evolved with signal *perception* (the four signal inputs) severed, while emission is left intact. This is the *deaf* (ablated) control; comparing evolved yield with and without perception isolates what communication is worth.

Oracle validation. Before trusting evolution, a verification that the world can actually reward communication is made through a hand-scripted oracle that emits while on food and climbs the signal gradient otherwise. Against an identical but *mute* oracle it raised group yield from a median 118.0 to 189.5 items, a +61% effect (Mann–Whitney $p \approx 4.5 \times 10^{-101}$, $n = 500$ groups).

Therefore, we can safely state that any failure to evolve communication is an evolutionary result, not an environment bug.

III. METHODS

A. Neural Networks and Implementation

Every agent is a fixed-topology network mapping the 9-D observation to its 3 outputs through one hidden layer of 16 tanh units; the move is tanh, the signal (a scalar $\in [0, 1]$) a sigmoid unit.

The two networks type differ in one crucial structural element:

- **Feedforward (FF):** memoryless MLP, $9 \rightarrow 16 \rightarrow 3$, 211 weights.
- **Recurrent (RNN):** Elman network [3], one of the simplest Neural Network architecture supporting memory, with $h_t = \tanh(W_{in}x_t + W_{rec}h_{t-1} + b)$, thus adding a 16×16 recurrent block (467 weights). The hidden state is zeroed for each episode and should give a receiver the memory to "keep heading toward a call heard a moment ago" and discern between long-lasting and possibly continuous signals and new ones.

A minimal network keeps the genome short enough to evolve at this scale while also guarding against the worry that any emergent communication is merely the consequence of an excessively complicated controller.

The simulator is pure NumPy: every agent in every group is advanced together with batched `einsum` forward passes and a fully vectorised toroidal world, and the full sweep is run across CPU cores.

B. Evolutionary algorithm

The network is initialized with random weights generated from a Gaussian distribution $\mathcal{N}(0, 0.5)$.

Each new generation is produced by a custom generational GA:

- The 4 best genomes are copied unchanged in the next generation (*elitism*) to ensure that the best coordinators are not lost;
- For the remaining 246, a *tournament* with ($k=3$) draws two parent pools. For each couple of parents, a 50/50 *crossover* is carried out with probability 0.3. Each gene is then *mutated* with Gaussian noise ($\sigma=0.1$) at a high per-gene rate of 0.9. The high-rate, small-step mutation is deliberate: many small perturbations explore the weight space more reliably than rare large jumps.

The experiments are performed with population 250 and 10 seeds per condition to ensure robustness. The feedforward network runs use 600 generations, by which point fitness had already plateaued; while the RNN runs are extended to 1000 generations, as 600 were not sufficient to reach the plateau.

C. Levels of selection and co-evolution

Both architectures are evolved under three different group formations, each with a different fitness target:

- **Colony:** each group consists of 8 clones of one genome (relatedness is 1); fitness is the whole group’s yield.
- **Individual:** a focal genome is placed among $G-1$ groupmates drawn at random from the population with the possibility of repetitions; fitness is the focal agent’s *own* intake.
- **Hybrid:** mixed groups in the individual case, but fitness is computed as α (personal) + $(1-\alpha)$ (group) with $\alpha=0.5$.

A form of intraspecific co-evolution arises in the two *mixed* regimes. There the non-focal groupmates are re-sampled from the current population every generation (averaged over `eval_repeats= 2` independent draws), so a genome’s fitness is *contextual*: it depends on the behaviour of other, distinct genomes that are themselves evolving, rather than on a fixed environment. *Colony* is not co-evolutionary in this sense, as every group member is a clone of the genome being scored, so its fitness depends only on itself.

The two co-evolutionary regimes differ in incentive alignment, and only the hybrid one is genuinely cooperative. Under *individual* selection a genome is rewarded solely for its own intake while its groupmates compete for the same food, and it is never credited for the groups in which it merely appears as a sampled partner; there is thus no pressure to help others, making this closer to *competitive* co-evolution.

Increasing `eval_repeats` would further reduce the noise of the random draws, but at a significant cost in the number of full evaluation passes.

The a-priori hypothesis was that colony selection would build cooperation and put the channel to use, individual selection would neglect or abuse it, and the hybrid would fall between the two.

IV. MEASURING COMMUNICATION

Measuring communication emergence was the hardest methodological choice. In fact, because fitness is scored purely on food yield, the setup contains no built-in indicator of whether effective communication has actually emerged.

A first attempt relied on mutual information, $I(\text{signal}; \text{on_food})$, an entropy-based measure of how much knowing the emitted signal reduces uncertainty about whether the agent is currently on food. However, this proved *insufficient*: the signal neuron shares hidden units with the food sensors, so the emitted scalar correlates with food even on the ablated setup, where nobody is listening. High MI can therefore arise even with no communication.

Two changes addressed this. First, every MI value is reported as *channel-on minus deaf*: only the information that appears

because listeners are present can count as communication. Second, instead of asking only about food, the signal of each seed’s best genome is scored against several candidate referents: `on_food` (same as above); `neighbors`, the number of groupmates within a local radius (0.1), on the intuition that agents might signal not about food itself but about local crowding, since clustering too tightly is a poor foraging strategy; and `recent_intake`, which was discarded as almost perfectly collinear with `on_food`.

Even this more complete set of indicators is only a lower bound on communication, but it appears to be enough to tell apart a signal that merely happens to be informative from one that represents actual communication.

V. RESULTS

A. Foraging yield over generations

All conditions learn to forage (Fig. 1), yet two patterns in particular stand out. First, **the recurrent agents reach substantially higher yields** than the feedforward ones under colony and individual selection (median final group yield 194.0 and 192.1 for the RNN at 1000 generations vs. 139.7 and 136.1 for the FF at 600). This advantage, however, does *not* come a communication effect: as the next subsection shows, the RNN gains nothing from the channel. Memory simply yields a better solo foraging policy. Second, the channel’s effect is visible directly in the curves (Fig.1): the FF *hybrid* channel-on curve (solid) sits well above its deaf control (dashed), whereas for the RNN the hybrid and colony *on* curves falls *below* their deaf control: a live channel actively hurts the recurrent agents in those regimes.

B. Functional value of the channel

The ablation makes the verdict clear (Fig. 2). **For the feedforward controller, the channel is beneficial in every regime:** +7.5% under colony ($p \approx 10^{-4}$ on the one-sided Mann–Whitney U test), +2.7% under individual ($p = 0.03$), and +24% under the hybrid ($p < 10^{-3}$). **For the recurrent controller, no regime shows a significant benefit:** the colony is negative (−5.1%), the individual gain is not significant (+3.0%, $p = 0.4$), and the hybrid is significantly *harmful* (−12.6%). This is the project’s headline: communication *did* emerge spontaneously, but only when the agents lacked the memory to do without it.

C. Level of selection

On the communication axis the level-of-selection hypothesis proved at least partly true: within the feedforward controller, colony selection extracts more value from the channel than individual selection (+7.5% vs. +2.7%), consistent with aligned interests favoring positive recruitment. The slight surprise is that the *hybrid* regime, not the pure colony, produced the strongest communication (+24%). A possible interpretation comes from the fact that under pure colony selection the group is clonal, so sender and receiver are the same genome and there is no distinct other to address; under pure individual selection there is a real receiver but no fitness incentive to help it.

TABLE I
MEDIAN MI: CHANNEL-ON, DEAF, AND THEIR DIFFERENCE (ONE-SIDED MANN-WHITNEY p , ON>DEAF).

		on	deaf	on–deaf (p)
<i>Referent: on_food</i>				
FF	colony	0.464	0.495	−0.031 (0.83)
FF	individual	0.453	0.474	−0.021 (0.79)
FF	hybrid	0.314	0.478	−0.164 (1.00)
RNN	colony	0.237	0.366	−0.129 (1.00)
RNN	individual	0.198	0.356	−0.157 (0.99)
RNN	hybrid	0.119	0.290	−0.171 (1.00)
<i>Referent: neighbors</i>				
FF	colony	0.048	0.010	+ 0.038 ($< 10^{-3}$)
FF	individual	0.022	0.009	+ 0.014 (0.02)
FF	hybrid	0.143	0.007	+ 0.136 (0.01)
RNN	colony	0.071	0.094	−0.024 (0.99)
RNN	individual	0.039	0.093	−0.054 (0.99)
RNN	hybrid	0.036	0.072	−0.036 (0.99)

The hybrid is the only regime that combines *both* a genuinely distinct receiver (mixed groups) *and* a partial reward for the group’s success, so both responding to others and emitting to help others are selected at once.

The recurrent controller shows no such pattern, because it does not use the channel in any regime.

D. What is the signal about?

The multi-referent analysis (Table I) resolves the apparent paradox of informative-but-useless signals. For the `on_food` referent, the channel-on MI never exceeds the deaf baseline in any condition (e.g. FF hybrid 0.314 on vs. 0.478 deaf). The signal’s information about food is entirely the “free” structural by-product anticipated in Sec. IV; by this metric one could wrongly conclude nothing communicative happens, or even that listening destroys information.

The `neighbors` referent displays a different story. Here the channel-on MI *exceeds* the deaf baseline, significantly and *only* for the feedforward controller, in all three regimes (colony +0.038, $p < 10^{-3}$; individual +0.014, $p = 0.02$; hybrid +0.136, $p = 0.01$). This excess is listener-dependent by construction, so it is genuine communication. Moreover, its magnitude across regimes (hybrid > colony > individual) matches the ranking of functional value. The content of the evolved feedforward signal is thus not “there is food here” as was initially expected, but something closer to “where I am relative to you”: the agents modulate emission with the local configuration of groupmates, which is what coordinates spacing and recruitment and what actually lifts yield.

The recurrent agents show no listener-dependent excess on any referent (Table I).

VI. TUNING DECISIONS

Hyper-parameters were fixed across conditions for a fair comparison rather than tuned per cell. 600 generations were where pilot yield curves plateaued. Mutation ($\sigma=0.1$, rate 0.9) balanced exploration against elite stability; hidden width 16

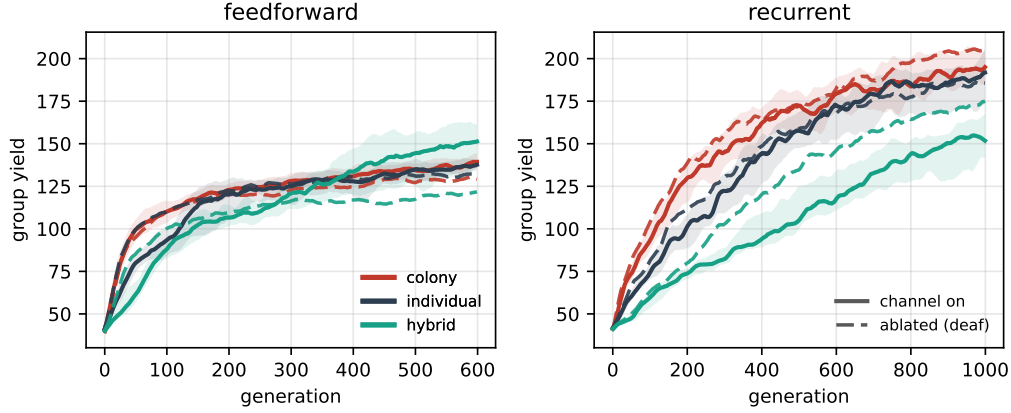


Fig. 1. Group yield vs. generation (median, 95% band, 10 seeds). The recurrent agents (right) forage higher overall, but the channel-on/ablated gap, the only part attributable to communication, favours the channel only for the feedforward agents (left), and is reversed for the recurrent hybrid.

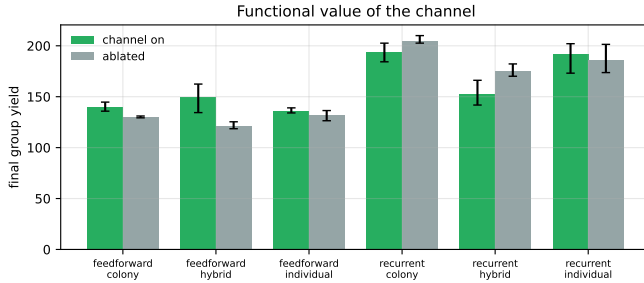


Fig. 2. Final group yield, channel-on vs. deaf (median, 95% CI). Feedforward agents (left three) consistently benefit; recurrent agents (right three) do not, and the recurrent hybrid is hurt by the channel.

was the smallest that kept the oracle-style policy representable without bloating the RNN genome. `eval_repeats=2` was a compromise to keep an acceptable cost and ten seeds were considered to give sufficient robustness to the results according to the Mann-Whitney test.

VII. DISCUSSION AND LIMITATIONS

MI does not tell the whole story. Even the improved metric cannot describe communication exhaustively. The deaf subtraction and the `neighbors` referent were necessary to see anything real, yet a signal could in principle carry function that no single scalar referent captures (like timing or sequence), and the recurrent results show that high informativeness can coexist with zero or negative function. We therefore treat the ablation yield as the operational definition of communication and MI as supporting, yet not defining, evidence.

Design choices. Three decisions were made that significantly shape the result:

- renewable (not finite) patches, which allow agents to stay on food and develop communication to attract (or repel) other agents and allow for communication evolution;
- non-exclusivity, since an agent alone can not deplete a food stash, this makes it less motivated to deter other

agents in an individual setting. This transformed the setting from hostile (agents rewarded if they sabotage rivals) to generally neutral;

- the focal-member construction of mixed groups, which keeps regimes compute-matched and differing in one factor.

Tuning. Because each full sweep was costly to simulate, most parameter choices were guided by short pilot runs and heuristics rather than an exhaustive search. It is therefore possible that better-performing or faster-converging settings exist, although the qualitative conclusions hold consistently across seeds and across the configurations that were tried.

VIII. CONCLUSION AND FUTURE WORK

Holding architecture topology fixed and never rewarding the channel, we found that informative, *beneficial* communication can emerge spontaneously in memoryless feedforward foragers, most strongly under a hybrid level of selection. By contrast, memory-equipped recurrent agents forage better while deriving no benefit from the communication channel, and in the hybrid regime are even harmed by it. Contrary to the initial hypothesis, rather than complementing and expanding the communication channel, the memory property removes the pressure to use the channel while allowing agents to learn better individual foraging strategies.

The mutual information results surprisingly show that the most effective communication emerged about the configuration of groupmates rather than being directly about food, which is why the $I(\text{signal}; \text{food})$ metric, high even when nobody listens, misses it. This suggests that more subtle variables related to movement and groupmates' positions might help to explain the evolution of communication even more. Future works could focus on a richer multi-dimensional signal and/or more complex architectures.

APPENDIX A

SIGNALLING BEHAVIOUR OF A CHAMPION

To make the metric concrete, Fig. 3 dissects the signal of the feedforward hybrid champion (best performing genome in a seed), the most communicative condition. Its emission has two distinct components. Panel (a) shows that the agent signals almost only when it is *on food*: the signal is high at zero distance and collapses past the sensing radius. This food cue is conspicuous, but, as argued in Sec. IV, it is no higher with listeners than without (Table I), so it reflects the shared input encoding rather than communication. Panel (b) on the other hand shows the genuinely communicative component: emission also grows with the number of nearby groupmates.

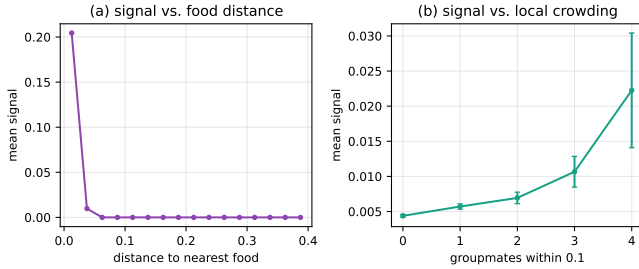


Fig. 3. Feedforward hybrid champion. (a) Mean emitted signal vs. distance to the nearest food item (b) Mean emitted signal vs. the number of groupmates within 0.1.

APPENDIX B

MI VISUALIZATION

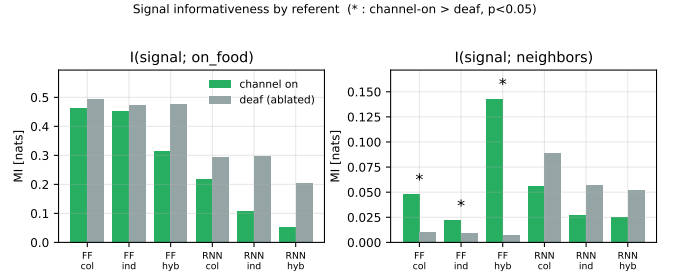


Fig. 4. Visualisation of Table I. *Left*: information about own food-state is as high or higher when deaf — it is “free”. *Right*: information about the configuration of groupmates exceeds the deaf baseline only for the feedforward agents (*: on>deaf, $p < 0.05$), and most strongly under the hybrid — the signature of real, function-bearing communication.

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